# Exit Ticket · reflect and retrieve

## Exit Ticket

In direct addressing, the operand always means the literal value to be used.

## Use key vocabulary from memory

1. Explain immediate in your own words.

2. Explain direct in your own words.

3. Explain indirect in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Lesson 126 · Addressing Modes: Immediate, Direct, Indirect and Indexed

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Which LMC instruction reads a mailbox value into the accumulator?

### Q2

In a BST, which side holds a smaller value?

### Q3

How many bits are in three bytes?

### Q4

What is a function parameter?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Immediate versus direct — LOAD #12 uses literal 12; LOAD 12 uses contents of memory[12].

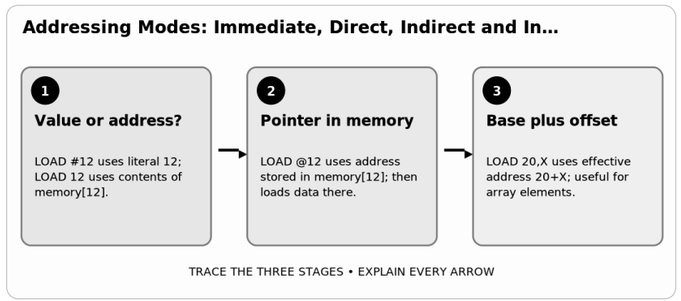
## Core idea 2

Indirect addressing — LOAD @12 uses address stored in memory[12]; then loads data there.

## Core idea 3

Indexed addressing — LOAD 20,X uses effective address 20+X; useful for array elements.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Explain indexed addressing for an array at base 50 with index 4 and discuss why this is not simply LMC LDA notation. [8]

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Define immediate and direct addressing with one example each. [6]

## Question 2

Memory[15]=29 and memory[29]=6. Give values loaded by direct LOAD 15 and indirect LOAD @15. [6]





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

immediate — Operand is a literal value.

direct — Operand gives the memory location holding a value.

indirect — Operand points to a location holding the effective address.

indexed — Base address plus index gives effective address.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

With memory[12]=40 and memory[40]=7, compare LOAD #12, LOAD 12 and LOAD @12.

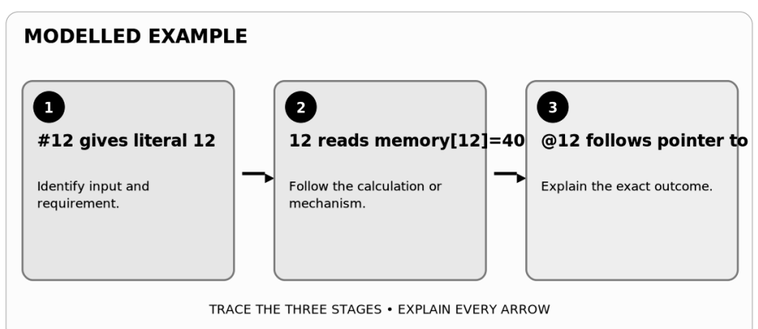
## Follow the worked method

1. #12 gives literal 12 — Predict the first step from the prompt.

2. 12 reads memory[12]=40 — Explain the intermediate result or decision.

3. @12 follows pointer to 7 — Using illustrative assembler conventions: immediate gives 12; direct reads 40; indirect follows memory[12]=40 then loads memory[40]=7.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Base 80, index 3
Predict the first step from the prompt.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.